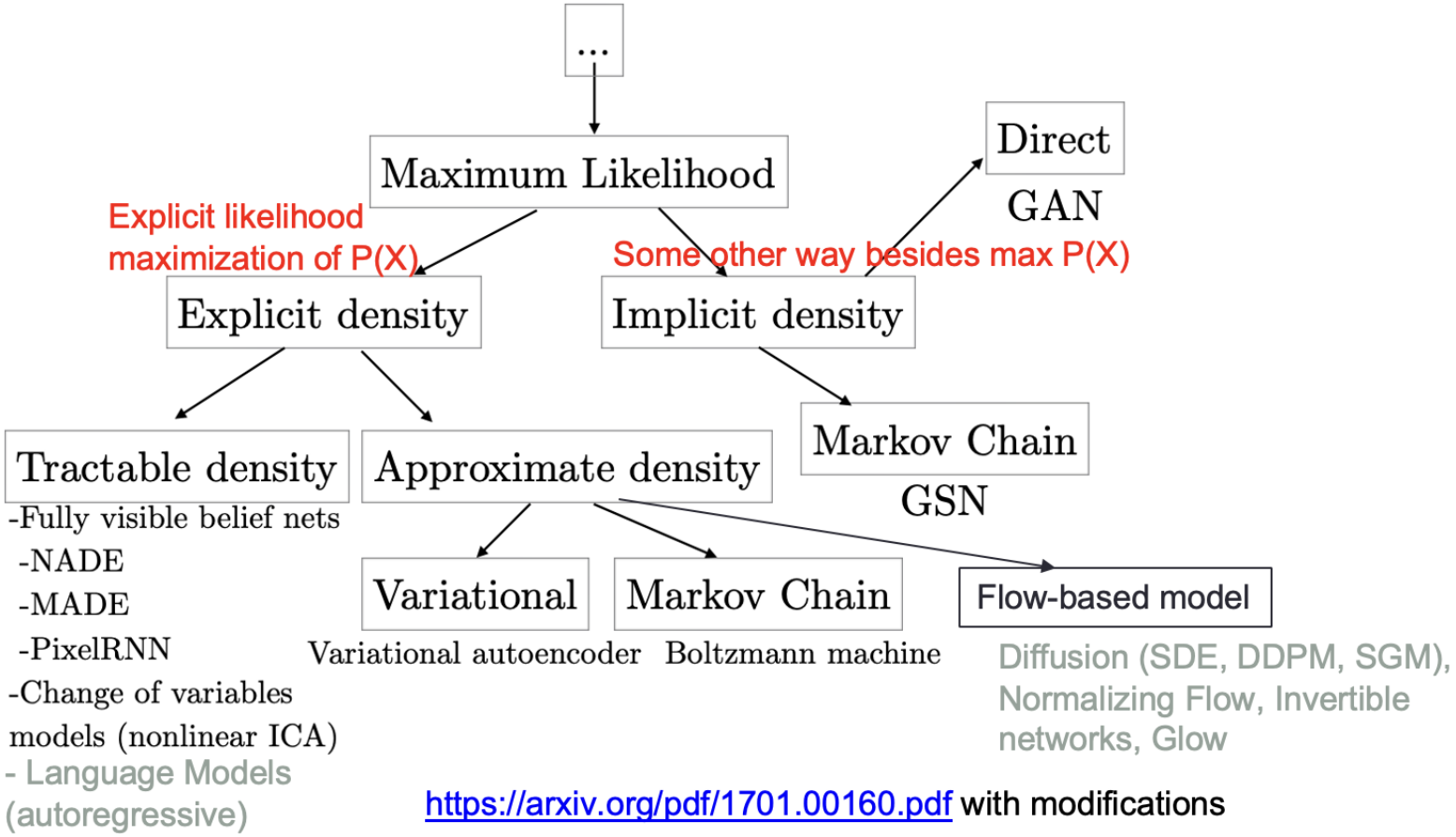
Deep Generative Models Part 2

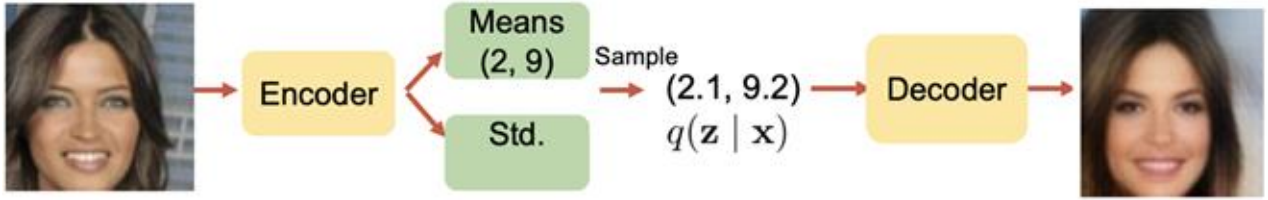
Flow, Diffusion

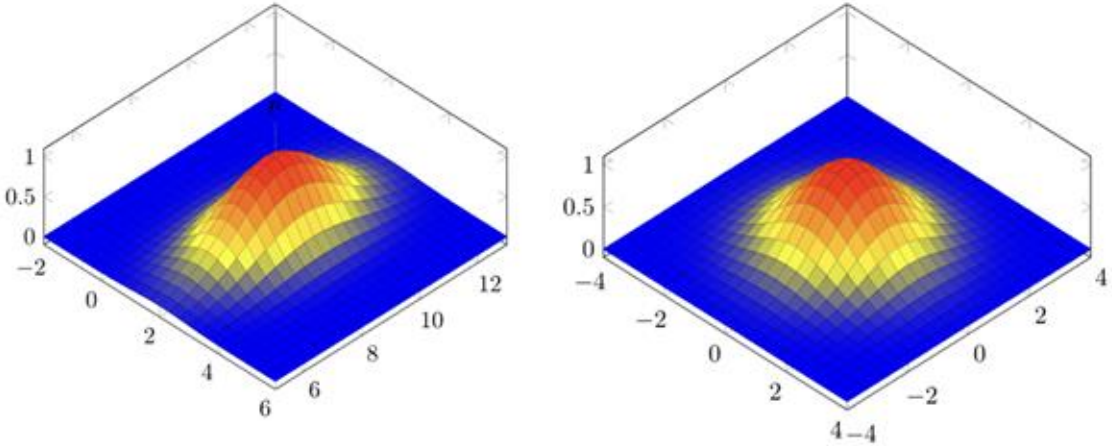
Classification based on training strategy



https://arxiv.org/pdf/1701.00160.pdf with modifications

VAE





$$q^*(\mathbf{z} \mid \mathbf{x}) = \underset{q}{\operatorname{argmin}}\, \mathrm{KL}\left[q(\mathbf{z} \mid \mathbf{x}) \| p(\mathbf{z} \mid \mathbf{x})\right]$$

ELBO

$q(\mathbf{z} \mid \mathbf{x})$ is forced to be close to $p(\mathbf{z}) = \mathcal{N}(0, I)$

$$\min_q \mathrm{KL}[q(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z})] - \mathbb{E}_q[\log p(\mathbf{x} \mid \mathbf{z})]$$

Force to be Gaussian ~L2

Effect of framing

$$D_{\mathrm{KL}}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log\left(\frac{P(x)}{Q(x)}\right)$$

1. Say we have a data distribution **p** that is a mixture of two 2D gaussians as shown below in red. We want to approximate this with one gaussian estimate **q** using KL-divergence. Which of the following three will result from optimizing $\mathbf{D}_{\mathrm{KL}}\ (p\ |\ |\ q)$
2. and which from $\mathbf{D}_{\mathrm{KL}}\ (q\ |\ |\ p)$?

$\mathbf{D}_{\mathrm{KL}}\ (q\ |\ |\ p)$ $\mathbf{D}_{\mathrm{KL}}\ (q\ |\ |\ p)$ $\mathbf{D}_{\mathrm{KL}}\ (p\ |\ |\ q)$

https://deeplearning.cs.cmu.edu/F23/document/slides/lec23.diffusion.updated.pdf

GAN

- Generator (Money Faker):
  - Maximize Y

$$\boxed{\min_G \mathbb{E}_{\mathbf{z}}[\log(1 - D(G(\mathbf{z}))]}$$

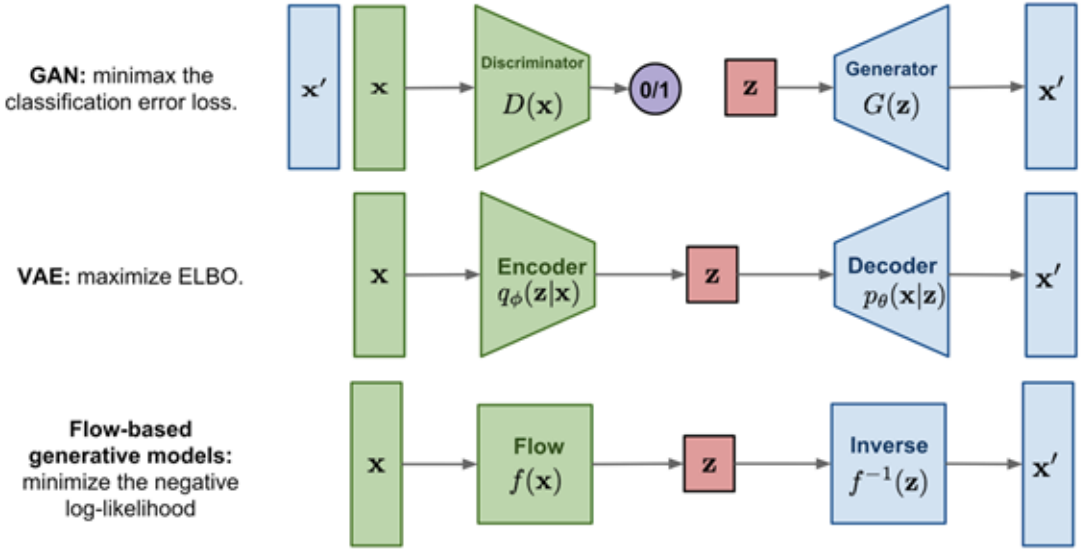
- Discriminator (Police):
  - For real images => Maximize Y
  - For generated images from the faker => Minimize Y

$$\boxed{\max_D \mathbb{E}_{\mathbf{z}}[\log(1 - D(G(\mathbf{z}))] + \mathbb{E}_{\mathbf{x}\sim p_{\text{data}}}[\log(D(\mathbf{x}))]}$$

This comes from binary cross entropy loss

Flow-based model

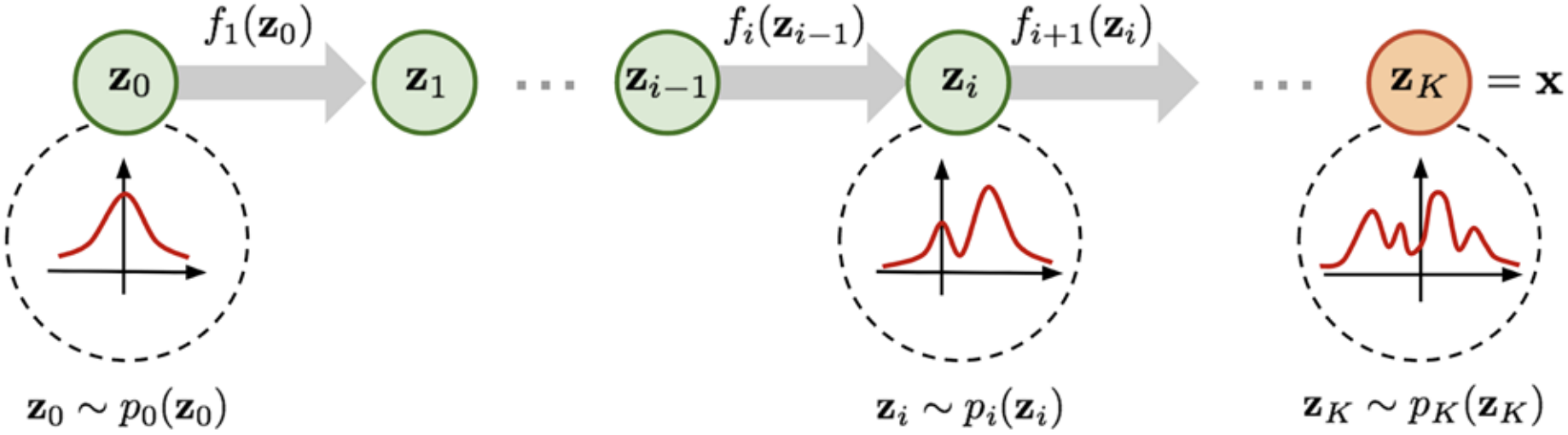
- Tweak distribution until you get the target distribution
- Deterministic reversible/invertable tweaks



https://lilianweng.github.io/posts/2018-10-13-flow-models/

https://lilianweng.github.io/posts/2018-10-13-flow-models/

Flow-based model



Noise

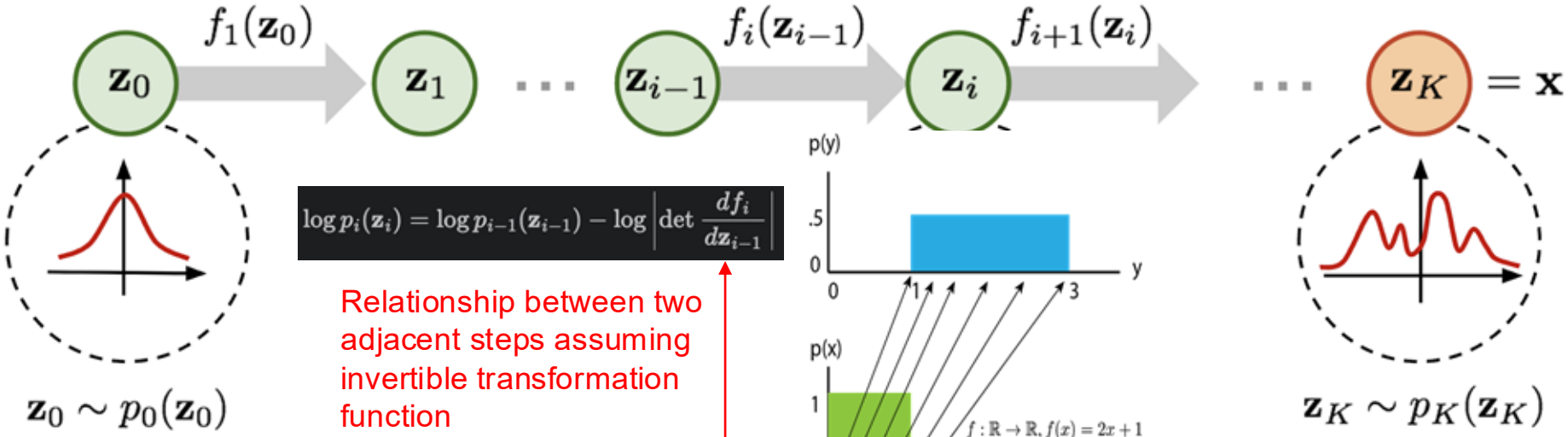
Target data

A flow-based model is a method that transforms a simple distribution to a complex one by successively applying a transformation function.

Here we are going to assume the transformation function is invertible.

https://lilianweng.github.io/posts/2018-10-13-flow-models/

Flow-based model



$$\log p_i(\mathbf{z}_i) = \log p_{i-1}(\mathbf{z}_{i-1}) - \log \left| \det \frac{df_i}{d\mathbf{z}_{i-1}} \right|$$

Relationship between two adjacent steps assuming invertible transformation function

Noise

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

The Jacobian matrix (matrix of first order partial derivatives)

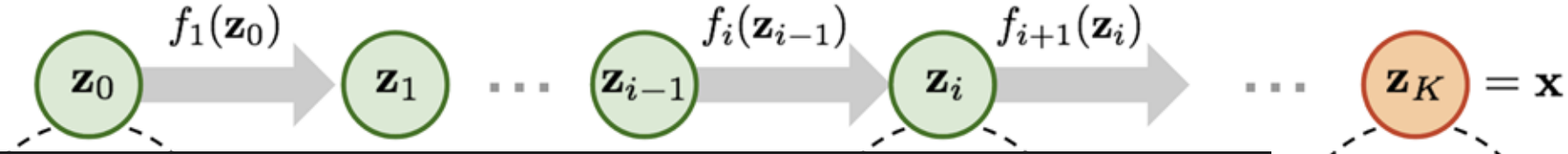
Target data

https://lilianweng.github.io/posts/2018-10-13-flow-models/

Flow-based model

$$\mathbf{J} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}$$

$$\log p_i(\mathbf{z}_i) = \log p_{i-1}(\mathbf{z}_{i-1}) - \log \left| \det \frac{df_i}{d\mathbf{z}_{i-1}} \right|$$

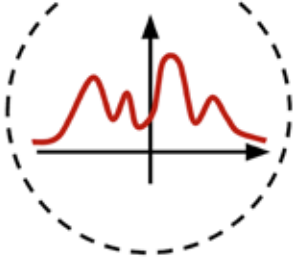


$$\mathbf{x} = \mathbf{z}_K = f_K \circ f_{K-1} \circ \cdots \circ f_1(\mathbf{z}_0)$$

$$\begin{aligned} \log p(\mathbf{x}) = \log \pi_K(\mathbf{z}_K) &= \log \pi_{K-1}(\mathbf{z}_{K-1}) - \log \left| \det \frac{df_K}{d\mathbf{z}_{K-1}} \right| \\ &= \log \pi_{K-2}(\mathbf{z}_{K-2}) - \log \left| \det \frac{df_{K-1}}{d\mathbf{z}_{K-2}} \right| - \log \left| \det \frac{df_K}{d\mathbf{z}_{K-1}} \right| \\ &= \dots \\ &= \log \pi_0(\mathbf{z}_0) - \sum_{i=1}^{K} \log \left| \det \frac{df_i}{d\mathbf{z}_{i-1}} \right| \end{aligned}$$

Normalizing flow

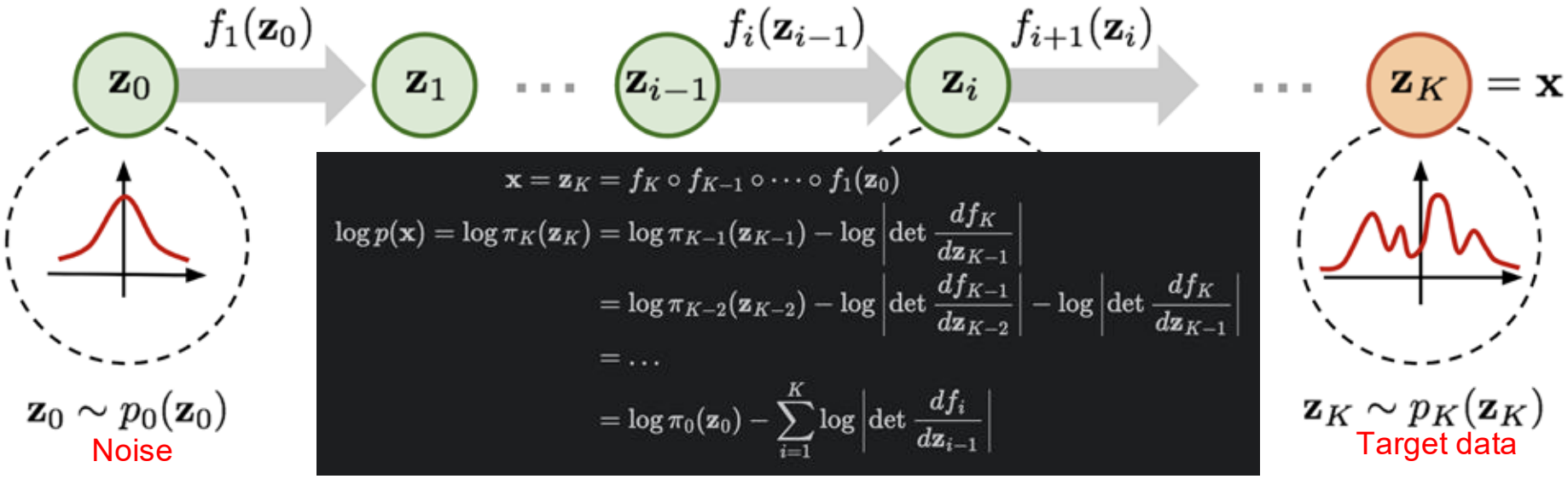
Now we can directly maximize log p(x)



$\mathbf{z}_K \sim p_K(\mathbf{z}_K)$

https://lilianweng.github.io/posts/2018-10-13-flow-models/

Flow-based model



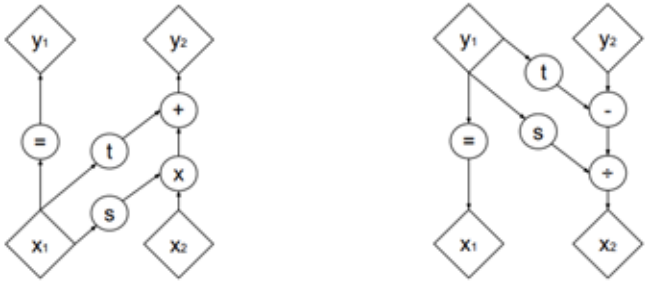
$$\mathbf{x} = \mathbf{z}_K = f_K \circ f_{K-1} \circ \cdots \circ f_1(\mathbf{z}_0)$$

$$\begin{aligned}
\log p(\mathbf{x}) = \log \pi_K(\mathbf{z}_K) &= \log \pi_{K-1}(\mathbf{z}_{K-1}) - \log \left| \det \frac{df_K}{d\mathbf{z}_{K-1}} \right| \\
&= \log \pi_{K-2}(\mathbf{z}_{K-2}) - \log \left| \det \frac{df_{K-1}}{d\mathbf{z}_{K-2}} \right| - \log \left| \det \frac{df_K}{d\mathbf{z}_{K-1}} \right| \\
&= \dots \\
&= \log \pi_0(\mathbf{z}_0) - \sum_{i=1}^{K} \log \left| \det \frac{df_i}{d\mathbf{z}_{i-1}} \right|
\end{aligned}$$

Find f that maximizes the likelihood of the data
Constrain f so that it is easily invertable and Jacobian easy to compute
f is a neural network (needs networks that's easily invertable by design)

Example of an invertible layer

- RealNVP



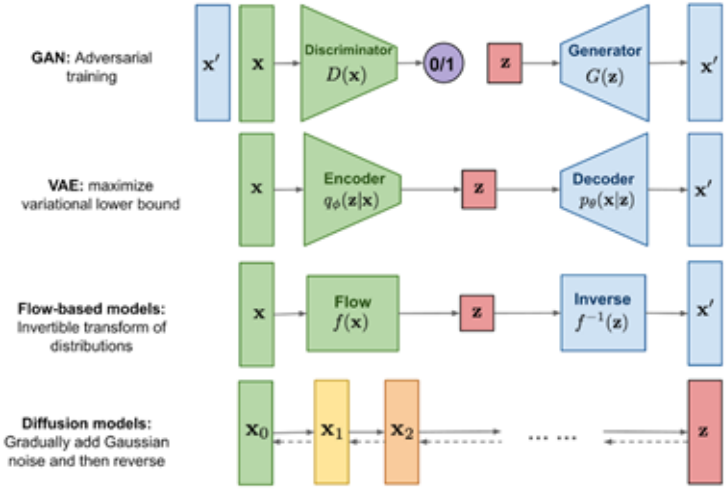
(a) Forward propagation (b) Inverse propagation

Figure 2: Computational graphs for forward and inverse propagation. A coupling layer applies a simple invertible transformation consisting of scaling followed by addition of a constant offset to one part $\mathbf{x}_2$ of the input vector conditioned on the remaining part of the input vector $\mathbf{x}_1$. Because of its simple nature, this transformation is both easily invertible and possesses a tractable determinant. However, the conditional nature of this transformation, captured by the functions s and t, significantly increase the flexibility of this otherwise weak function. The forward and inverse propagation operations have identical computational cost.

https://arxiv.org/abs/1605.08803

Diffusion Model

- Similar to flow
  - No longer constrained to revertible operations
  - Add noise to image until you get just noise



https://lilianweng.github.io/posts/2021-07-11-diffusion-models/

Lecture Plan

1. Why Diffusion? (Motivation)
2. What is Diffusion?
3. Types of Diffusion
    a. Probabilistic Model (Focus on this one)
    b. Stochastic Differential Equations
4. What is the problem of Diffusion? (Trends)
    a. Stable Diffusion
    b. Knowledge Distillation
5. Parameter Efficient Transfer Learning
    a. LORA
    b. Adapters

Benefits of Diffusion

- A promising result, high-quality image with fine details and realistic textures.
- Stable training, compared to GAN

Application

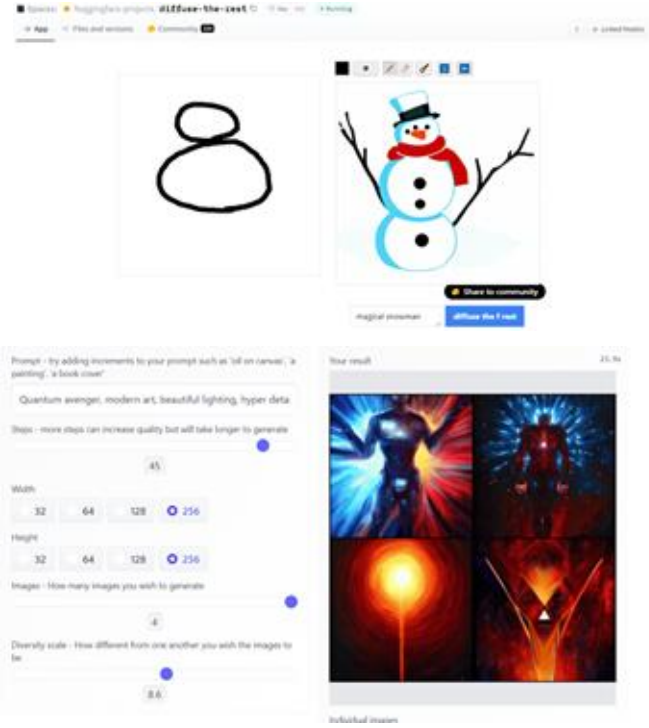


Image Generation

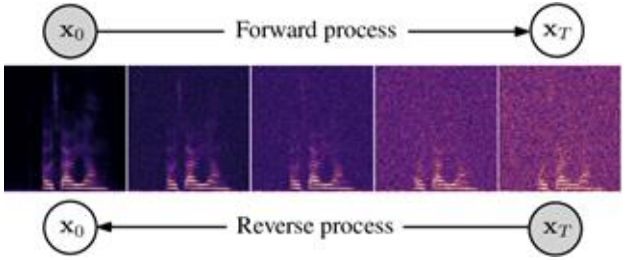
Image Restoration

Image Super Resolution

Beyond Image

3D Generation

Motion Generation



Speech Enhancement

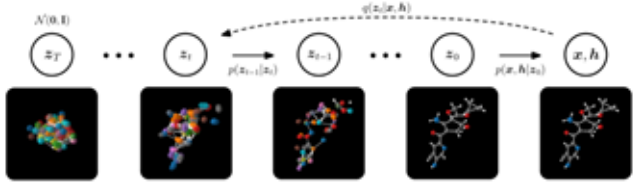


Figure 2. Overview of the Equivariant Diffusion Model. To generate molecules, coordinates $\boldsymbol{x}$ and features $\boldsymbol{h}$ are generated by denoising variables $\boldsymbol{z}_t$ starting from standard normal noise $\boldsymbol{z}_T$. This is achieved by sampling from the distributions $p(\boldsymbol{z}_{t-1}|\boldsymbol{z}_t)$ iteratively. To train the model, noise is added to a datapoint $\boldsymbol{x}, \boldsymbol{h}$ using $q(\boldsymbol{z}_t|\boldsymbol{x}, \boldsymbol{h})$ for the step t of interest, which the network then learns to denoise.

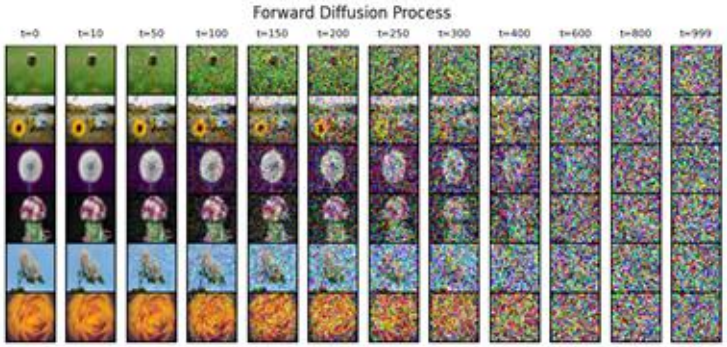
Graph Generation

What is Diffusion?

The concept of diffusion was adopted from non-equilibrium thermodynamics.

Diffusion model comprises of 2 process:

1. Forward Process (adding noise)
    a. Gradually insert gaussian noise into a clean image, until it becomes gaussian noise.
    b. Objective: Transform a complex distribution (a clean image) into a simple distribution (such as Gaussian distribution)
2. Reversed Process (model learns this part)
    a. Gradually remove gaussian noise, until it becomes a clean image.
    b. Objective: Generate a new image from sampled noise.



How to model diffusion process?

Diffusion process can be classify into two categories. (Ultimately they are equivalent)

1. Probabilistic Model
   a. Exploit Markov Chain process to diffusion process (repeatedly insert noise)
   b. Example: DDPM, DDIM

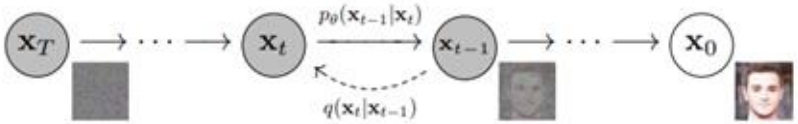
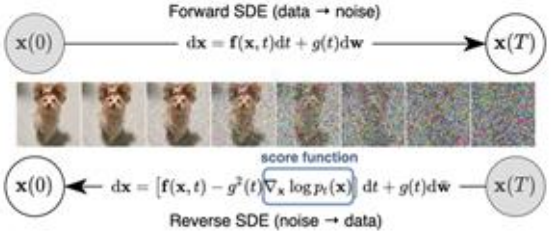


Figure 2: The directed graphical model considered in this work.

1. Stochastic Differential Equation
   a. Employ Stochastic Differential Equation to further extend the discrete process into continuous process
   b. Example: Score-based Diffusion model



Denoising Diffusion Probabilistic Model

Denoising Diffusion Probabilistic Models (DDPM)

Formulate the forward process with Markov Chain process

- Forward Process
  - First, we define the probability of x_t given x_{t-1} such that we can sample forward process with a mathematical process.

$$x_i = \sqrt{1-\beta_i}x_{i-1} + \sqrt{\beta_i}\epsilon_{i-1} \qquad i = 1, \cdots, N$$

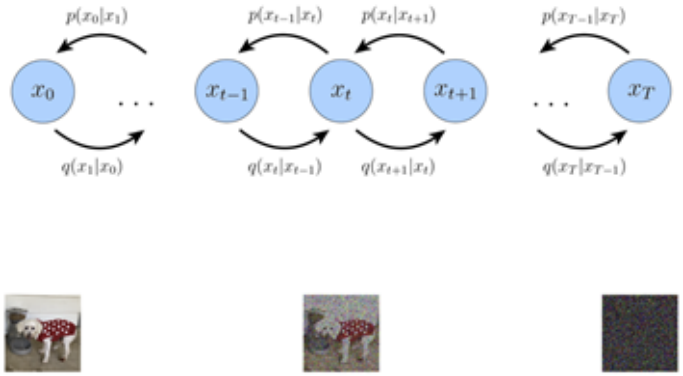
$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t; \qquad \epsilon_t \sim \mathcal{N}(0, \mathbb{I})$$

  - Given α_t is a decreasing sequence.

$$q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\mathbb{I})$$

  - From markov chain property, we can further extend the probability of $\{x_1, \ldots x_T\}$ given x_0

$$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1})$$



https://arxiv.org/abs/2006.11239 **Denoising Diffusion Probabilistic Models 2020**

Denoising Diffusion Probabilistic Models (DDPM)

What's the benefits of defining the forward process like this? $q(x_t|x_{t-1}) = \mathcal{N}(x_t; \sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\mathbb{I})$

- It allow us to sample $\mathbf{x}_t$ from $\mathbf{x}_0$, using the following equation

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_{t-1}, (1-\bar{\alpha}_t)\mathbb{I}) \qquad \bar{\alpha}_t = \prod_{t'=1}^{t'=t} \alpha'_t$$

- Proof

$$x_t = \sqrt{\alpha_t}x_{t-1} + \sqrt{1-\alpha_t}\epsilon_t; \qquad \epsilon_t \sim \mathcal{N}(0, \mathbb{I})$$

$$x_t = \sqrt{\alpha_t}(\sqrt{\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_{t-1}}\epsilon_{t-1}) + \sqrt{1-\alpha_t}\epsilon_t$$

$$x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{\alpha_t(1-\alpha_{t-1})}\epsilon_{t-1} + \sqrt{1-\alpha_t}\epsilon_t$$

$$x_t = \sqrt{\alpha_t\alpha_{t-1}}x_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\epsilon'_t; \qquad \epsilon'_t \sim \mathcal{N}(0, \mathbb{I})$$

We combine ϵ_{t-1} and ϵ_t into ϵ'_t by using the following theorem.

1. If a is a constant and $x \sim \mathcal{N}(\mu, \sigma^2)$, then $ax \sim \mathcal{N}(a\mu, (a\sigma)^2)$
2. If $x_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $x_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, then $x_1 + x_2 \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$

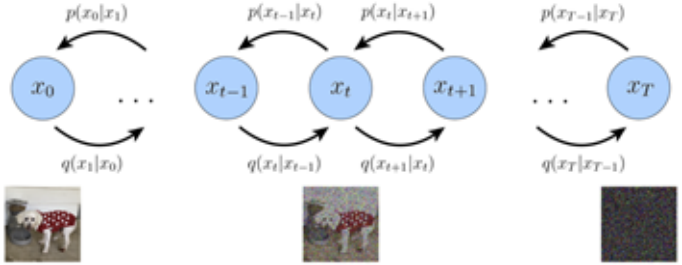
Denoising Diffusion Probabilistic Models (DDPM)

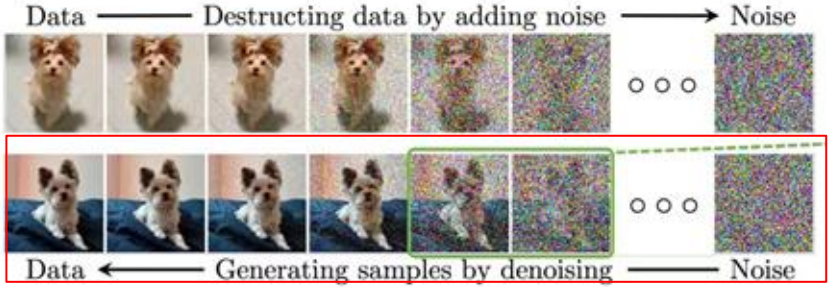
Reverse process is a generative process which we want to learn using deep learning model.

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

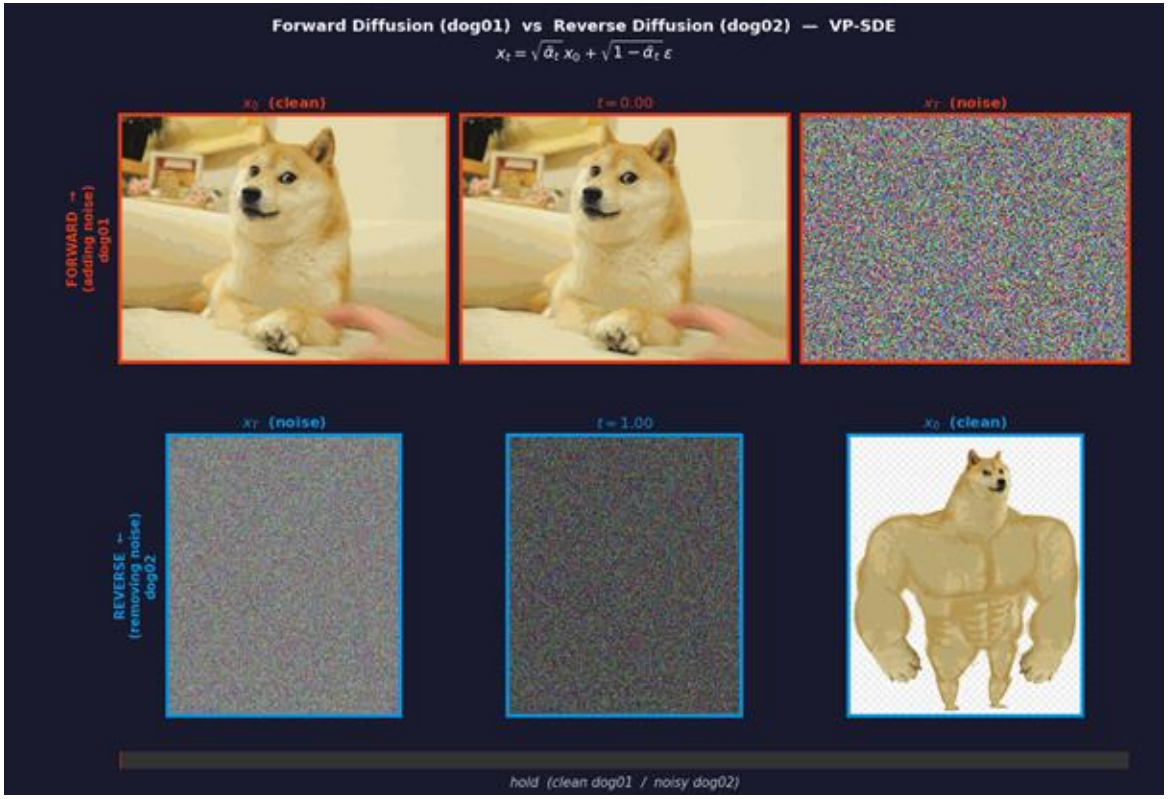
To sample image from reverse process:

1. Sample a noise from Gaussian distribution, $\mathbf{x_T}$
2. Deploy our deep learning model to repeatedly remove the noise





Denoising Diffusion Probabilistic Models (DDPM)



Denoising Diffusion Probabilistic Models (DDPM)

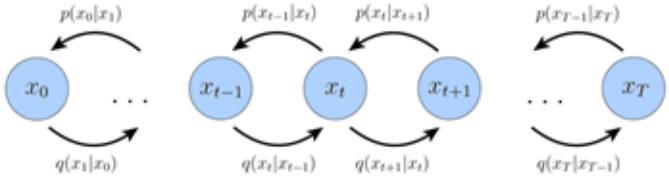
In a typical generative model, we want to learn the distribution of the data (clean images).

However, it is intractable to compute due to

- Infeasible to compute $p(x_0)$, due to the enormous space of $x_{1:T}$
- Unknown data distribution, $q(x_0)$

$$p(x_0) = \int p(x_0|x_{1:T})p(x_{1:T})dx_{1:T}$$

Fortunately, we have a technique from VAE, Evidence Lower-Bound (ELBO)



Consider $D_{\mathrm{KL}}(q(x_{1:T}|x_0)||p(x_{1:T}|x_0))$

$$= \mathbb{E}_{x_{1:T}\sim q}[\log q(x_{1:T}|x_0) - \log p(x_{1:T}|x_0)]$$

$$= \mathbb{E}_{x_{1:T}\sim q}[\log q(x_{1:T}|x_0) - \log \frac{p(x_{0:T})}{p(x_0)}]$$

$$= \mathbb{E}_{x_{1:T}\sim q}[\log q(x_{1:T}|x_0) - \log p(x_{0:T})] + \mathbb{E}_{x_{1:T}\sim q}[\log p(x_0)]$$

KL always >= 0

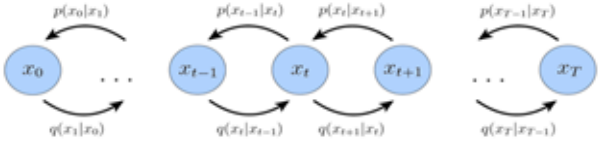
Therefore $\boxed{\mathbb{E}_{x_0\sim q(x_0)}[-\log p(x_0)]} < \boxed{\mathbb{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_{1:T}|x_0) - \log p(x_{0:T})]}$ Tractable to compute

Data Distribution

Denoising Diffusion Probabilistic Models (DDPM)

For explainability, loss function is splitted into 3 terms. $\mathbb{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_{1:T}|x_0) - \log p(x_{0:T})]$

To decompose this equation, we use three equations.



- Forward process is a markov process

$$q(x_{1:T}|x_0) = \prod_{t=1}^{T} q(x_t|x_{t-1}) \qquad q(x_t|x_{t-1}) = q(x_t|x_{t-1}, x_0)$$

- Reverse process is also a markov process

$$p(x_{0:T}) = p(x_T)\prod_{t=1}^{T} p(x_{t-1}|x_t)$$

See appendix

We learn this term

$$\mathbb{E}_{x_{0:T}\sim q(x_{0:T})}[\underbrace{D_{\mathrm{KL}}(q(x_T|x_0)||p(x_T))}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))}_{L_{t-1}} - \underbrace{\log p(x_0|x_1)}_{L_0}]$$

Prior Matching of starting point

Denoising

Reconstruction at first step

Denoising Diffusion Probabilistic Models (DDPM)

Our deep learning model is used to predict mean of the reverse process. For simplicity we set variance equal to untrained time dependent constant.

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

L_T Loss $\quad D_{\mathrm{KL}}(q(x_T|x_0)||p(x_T))$

- This term is a constant and can be ignored because the scheduler (α_t) has no learnable parameters and prior sampling is sampled from Gaussian noise.

L_{t-1} Loss $\quad D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))$

- We need $q(x_{t-1} \mid x_t, x_0)$ to compute this

L_0 Loss $\quad \log p(x_0|x_1)$

- Probability of x_0 given x_1 is Normal distribution with predicted mean. $\log(p(x_0|x_1)) = \frac{1}{2\sigma_1^2}(x_0 - \mu_\theta(x_1, 1))^2 + C$

Denoising Diffusion Probabilistic Models (DDPM)

Fortunately $\mathbf{L_{t-1}}$ doesn't require the actual probability of reverse process which is intractable and it only require to calculate probability of reverse process given $\mathbf{x_0}$.

This term is much simpler to calculate.

$$D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))$$

$$\begin{aligned} q(x_{t-1}|x_t, x_0) &= \frac{q(x_{t-1}, x_t, x_0)}{q(x_t, x_0)} \\ &= \frac{q(x_t|x_{t-1}, x_0)q(x_{t-1}, x_0)}{q(x_t, x_0)} \\ &= \frac{q(x_t|x_{t-1})q(x_{t-1}|x_0)}{q(x_t|x_0)} \end{aligned}$$

See appendix

$$q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t \mathbb{I})$$

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t} x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t} x_0$$

$$\tilde{\beta}_t = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

$$D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t)) = \mathbb{E}_q[\log \frac{\sqrt{\Sigma_\theta}}{\sqrt{\tilde{\beta}_t}} \exp(-\frac{1}{2}(\frac{(x_{t-1}-\tilde{\mu}_t)^2}{\tilde{\beta}_t} - \frac{(x_{t-1}-\mu_\theta)^2}{\Sigma_\theta})]$$

$$D_{\mathrm{KL}}(q(x_{t-1}|x_t,x_0)||p_\theta(x_{t-1}|x_t)) = \mathbb{E}_q[\log \frac{\sqrt{\Sigma_\theta}}{\sqrt{\tilde{\beta}_t}} \exp(-\frac{1}{2}(\frac{(x_{t-1}-\tilde{\mu}_t)^2}{\tilde{\beta}_t} - \frac{(x_{t-1}-\mu_\theta)^2}{\Sigma_\theta})]$$

Denoising Diffusion Probabilistic Models (DDPM)

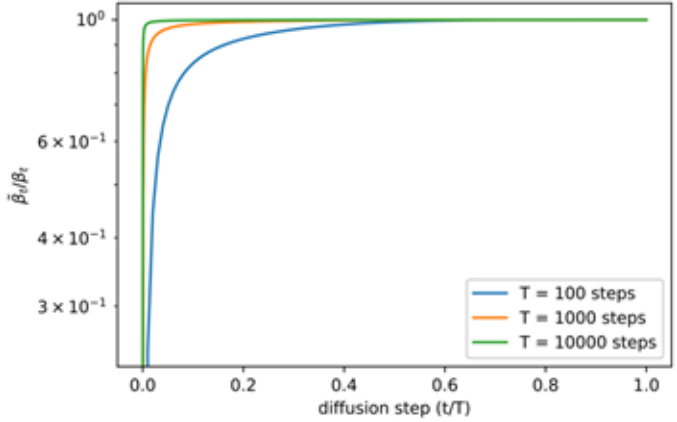
$$\tilde{\beta}_t = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

The actual variance of reverse process. $q(x_{t-1}|x_t)$

- Upper-bound: The variance of forward process $q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{1-\beta_t}x_{t-1}, \beta_t\mathbb{I})$
  - Optimal for $x_0 \sim \mathcal{N}(0,\mathbb{I})$
- Lower-bound: The variance of reverse process given x_0 $\quad q(x_{t-1}|x_t,x_0) = \mathcal{N}(x_{t-1};\tilde{\mu}_t,\tilde{\beta}_t\mathbb{I})$
  - Optimal for x_0 determinedly set to one point.
  - It is an extreme case for the reverse process because we don't know x_0.

Based on the paper, Σ_θ was set to its upper bound.

Note that the range of upper-bound and lower-bound is extremely small when diffusion step is large.



Denoising Diffusion Probabilistic Models (DDPM)

In summary, our loss function $\mathbb{E}_{x_{0:T}\sim q(x_{0:T})}[\underbrace{D_{\mathrm{KL}}(q(x_T|x_0)||p(x_T))}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t))}_{L_{t-1}} - \underbrace{\log p(x_0|x_1)}_{L_0}]$

L_T Loss

- Constant with respect to **theta** (reverse process).

L_{t-1} Loss

$$D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p_\theta(x_{t-1}|x_t)) = \mathbb{E}_q[\frac{1}{2\tilde{\beta}_t}(\tilde{\mu}_t - \mu_\theta)^2] + C$$

L_0 Loss

$$\mathbb{E}_q[-\log p_\theta(x_0|x_1)] = \mathbb{E}_q[\frac{1}{2\tilde{\beta}_1}(x_0 - \mu_\theta)^2] + C$$

Denoising Diffusion Probabilistic Models (DDPM)

Besides modeling the mean of the reverse process, we can further parameterize the model to predict the noise instead.

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}x_0$$
$$= \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_t)$$
$$= \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_t)$$

$$q(x_t|x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I})$$
$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon_t$$
$$x_0 = \frac{1}{\sqrt{\bar{\alpha}_t}}(x_t - \sqrt{1-\bar{\alpha}_t}\epsilon_t)$$

From the above equation, we parameterize the model to predict noise.

$$p_\theta(x_{t-1}|x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t))$$

$$\mu_\theta = \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta)$$
$$\Sigma_\theta = \tilde{\beta}_t$$

Modeling the noise leads to easier learning*. It also has nice theoretical interpretations (see paper).

* Not always, according to my experience

Denoising Diffusion Probabilistic Models (DDPM)

Substitute the equations from the previous page into Loss function

- L_{t-1} Loss: $\mathbf{E}_q[\frac{1}{2\tilde{\beta}_t}(\tilde{\mu}_t - \mu_\theta)^2] = \mathbf{E}_q[\frac{\beta_t^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}(\epsilon_t - \epsilon_\theta)^2]$

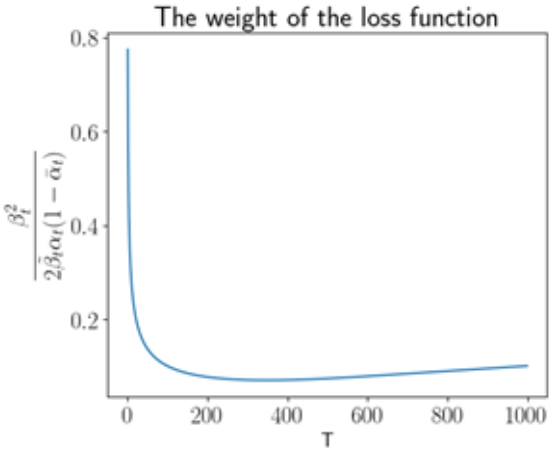
- L_0 Loss: $\mathbf{E}_q[\frac{1}{2\tilde{\beta}_1}(x_0 - \mu_\theta)^2] = \mathbf{E}_q[\frac{\beta_1^2}{2\tilde{\beta}_1\alpha_1(1-\bar{\alpha}_1)}(\epsilon_1 - \epsilon_\theta)^2]$

From our parameterization, it shows that both L_{t-1} and L_0 have the same form and can be merged.

$$\mathbf{E}_{t\sim\mathrm{Uniform}(0,T),x_0\sim q,\epsilon_t\sim\mathcal{N}(0,\mathbb{I})}[\boxed{\frac{\beta_t^2}{2\tilde{\beta}_t\alpha_t(1-\bar{\alpha}_t)}}(\epsilon_t - \epsilon_\theta)^2]$$

Can be ignored

The paper also remarks that the weight can be omitted to encourage the model to focus on more challenging denoising tasks at larger time steps (higher t terms).



Denoising Diffusion Probabilistic Models (DDPM)

Algorithm 1 Training

1: **repeat**
2: $x_0 \sim q(x_0)$
3: $t \sim \mathrm{Uniform}(\{1, \ldots, T\})$
4: $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
5: $x_t \leftarrow \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$
6: Take gradient descent step on $\nabla_\theta ||\epsilon - \epsilon_\theta(x_t, t)||^2$
7: **until** converged

Algorithm 2 Sampling

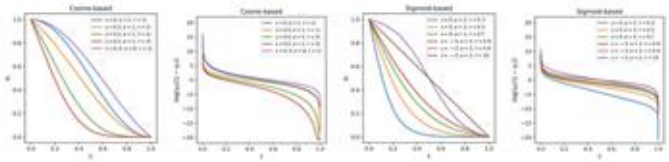
1: $x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
2: **for** $t = T, \ldots, 1$ **do**
3: $z \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ if $t > 1$, else $z = 0$
4: $x_{t-1} = \frac{1}{\sqrt{\alpha_t}}(x_t - \frac{1-\alpha_t}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(x_t, t)) + \sigma_t z$
5: **end for**
6: **return** x_0

Noise schedule α_t

- In the DDPM paper
  - We chose the β_t schedule from a set of constant, linear, and quadratic schedules, all constrained so that $L_T \approx 0$. We set $T = 1000$ without a sweep, and we chose a linear schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$.

- It is found that Best noise schedule depends on the resolution of the image and the task

- Thus, it is important to find a proper noise schedule for your task



(a) Cosine (b) Cosine (logSNR) (c) Sigmoid (d) Sigmoid (logSNR)

Figure 3: Instantiations of noise schedule function $\gamma(t)$ and the corresponding logSNR. Adjusting hyper-parameters of cosine and sigmoid functions leads to different noise schedules.

Frechet Inception Distance (FID) inception score (IS)

Resolution	Method	FID	IS	Params (M)
64×64	ADM [3]	–	2.07	297
	CF-guidance [6]	1.55	66.0	–
	CDM [8]	1.48	66.0	–
	RIN [10] (patch size of 4, 300K updates)	**1.23**	**66.5**	281
	RIN+our strategy (patch size of 8, 150K updates)	2.04	55.8	**214**
128×128	ADM [3]	5.91	–	386
	ADM+guidance [3]	2.97	–	> 386
	CF-guidance[6]	**2.43**	**156.0**	–
	CDM[8]	3.51	128.0	1058
	RIN [10] (patch size of 4, 700K updates)	2.75	144.1	410
	RIN+our strategy (patch size of 8, 250K updates)	3.50	120.4	**215**
256×256	ADM [3]	10.94	100.9	553
	ADM+guidance [3]	4.59	–	>553
	CDM [8]	4.88	158.7	1953
	RIN [10] (patch size of 8, 700K updates)	4.51	161.0	410
	RIN+our strategy (patch size of 8, 250K updates)	**3.52**	**186.2**	**319**
512×512	ADM [3]	23.2	58.1	559
	ADM+guidance [3]	7.72	172.7	>559
	RIN+our strategy (patch size of 8, 1M updates)	**3.95**	**216**	**320**
768×768	RIN+our strategy (patch size of 8, 1M updates)	**5.60**	**196.2**	**408**
1024×1024	RIN+our strategy (patch size of 8, 910K updates)	**8.72**	**163.9**	**412**

https://arxiv.org/pdf/2301.10972.pdf 2023 On the Importance of Noise Scheduling for Diffusion Models

Inception Score

- A score that tries to capture fidelity and diversity of generated images
  - Use InceptionV3 (imagenet pretrained) to classify generated images. Good images should yield sharp softmax probabilities (low entropy per prediction).
  - Good generation should also have a good overall spread of predictions across multiple generations (high entropy across predictions)

Original form in paper

$$\ln IS(p_{gen}, p_{dis}) := \mathbb{E}_{x \sim p_{gen}} \left[D_{KL} \left(p_{dis}(\cdot|x) \| \mathbb{E}_{x \sim p_{gen}} [p_{dis}(\cdot|x)] \right) \right]$$

$$\ln IS(p_{gen}, p_{dis}) := H[\mathbb{E}_{x \sim p_{gen}} [p_{dis}(\cdot|x)]] - \mathbb{E}_{x \sim p_{gen}} [H[p_{dis}(\cdot|x)]]$$

Equivalent rewrite

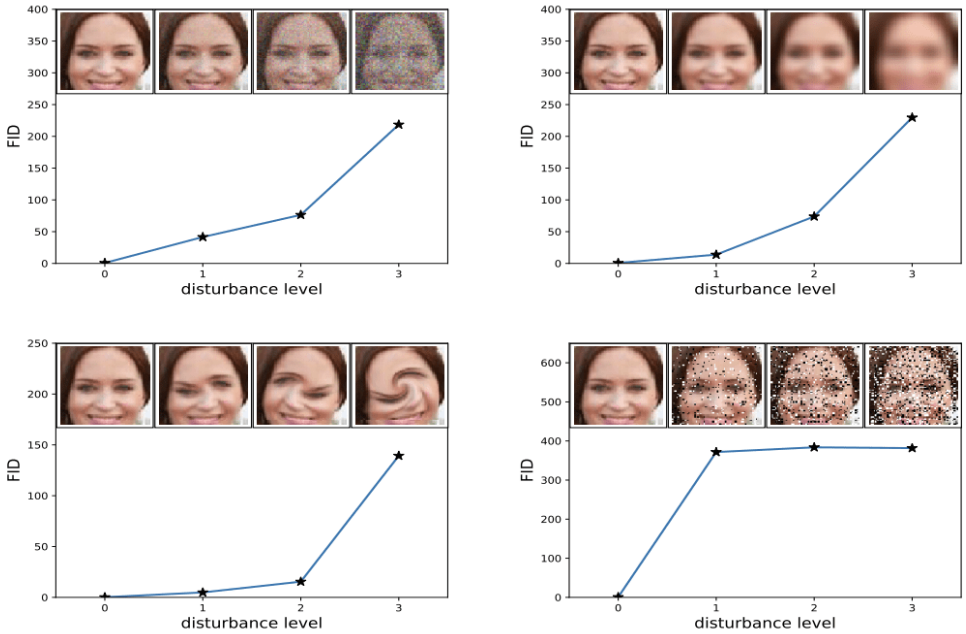
P_{dis} is inceptionv3 softmax value.

https://arxiv.org/abs/1606.03498

FID score

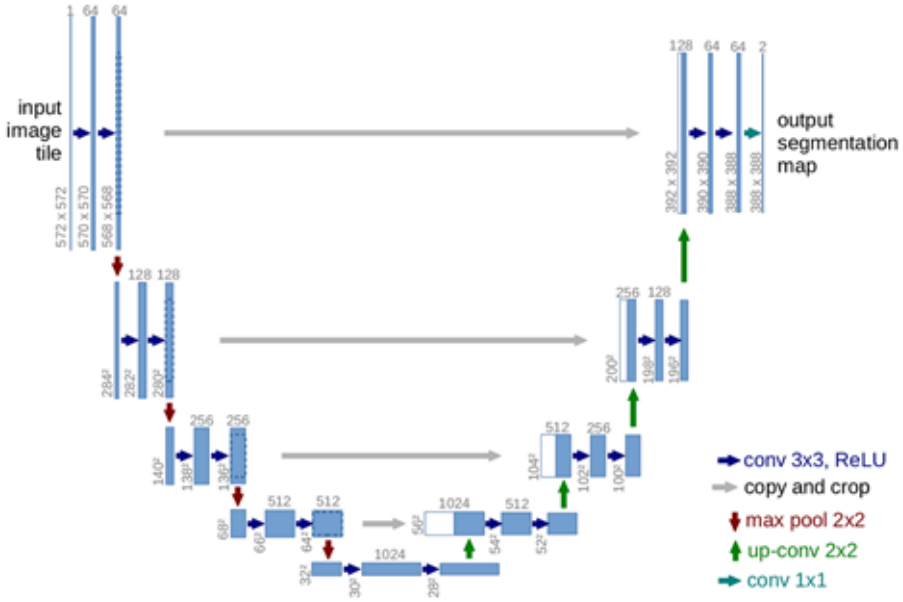
- Use features extracted from Inceptionv3 model (in the deeper layer). These features captures how high level features are found in the image.
- Fit Multivariate Guassians on training image m_{train}, Σ_{train} and generated images m_{gen}, Σ_{gen}
- Compute 2-Wasserstein Distance between Guassians

- Best (lowest) FID usually have best variety.
  Best (highest) IS usually have best quality.



https://arxiv.org/abs/1706.08500

Model architecture



Score-based Generative Modeling through Stochastic Differential Equation

Ordinary Differential Equations

- Ordinary differential equations (ODEs)
  - Trajectory (solution): $X : [0, T] \to \mathbb{R}^d,\ t \mapsto X_t$
  - Vector field (velocity in space): $u : \mathbb{R}^d \times [0, T] \to \mathbb{R}^d,\ (x, t) \mapsto u_t(x)$
  - Flow (specific instantiation if we start at x_0 at t=0): $\psi : \mathbb{R}^d \times [0, T] \to \mathbb{R}^d,\ (x_0, t) \mapsto \psi_t(x_0)$

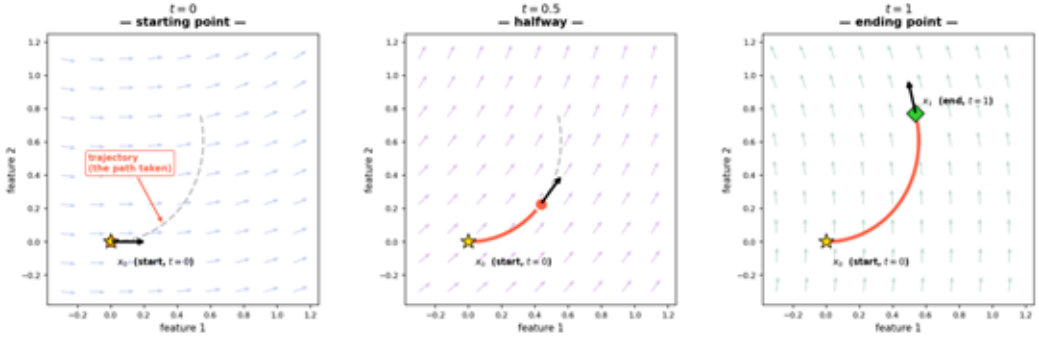
ODE

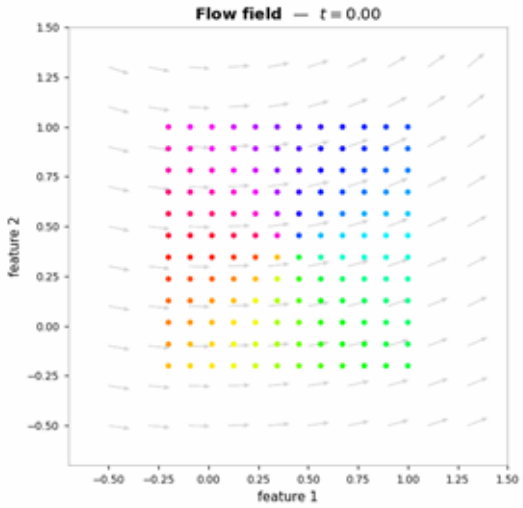
$$\frac{\mathrm{d}}{\mathrm{d}t} X_t = u_t(X_t)$$
$$X_0 = x_0$$

Flow ODE

A path if we start at $X_0 = x_0$ at t = 0

$$\frac{\mathrm{d}}{\mathrm{d}t} \psi_t(x_0) = u_t(\psi_t(x_0))$$
$$\psi_0(x_0) = x_0$$





Stochastic Differential Equations

- Stochastic Differential Equations (SDEs) move from **deterministic** trajectories in ODEs to **stochastic** trajectories.

$$X_t \text{ is a random variable}$$

- Brownian Motion (Wiener process: W)
  - Normal increments

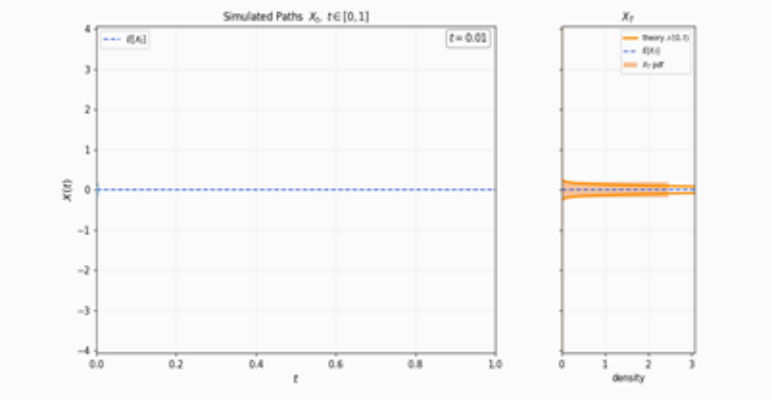
$$W_{t_2} - W_{t_1} \sim \mathcal{N}(0, (t_2 - t_1)I_d), \quad \text{for all } 0 \leq t_1 < t_2$$

  - Independent increments

$$W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}} \text{ are mutually indepedent}$$

$$\text{for } 0 \leq t_0 < t_1 < \dots < t_n$$

$$\Rightarrow W_{t+h} = W_t + \sqrt{h}\epsilon_t, \quad \epsilon_t \sim \mathcal{N}(0, I_d)$$



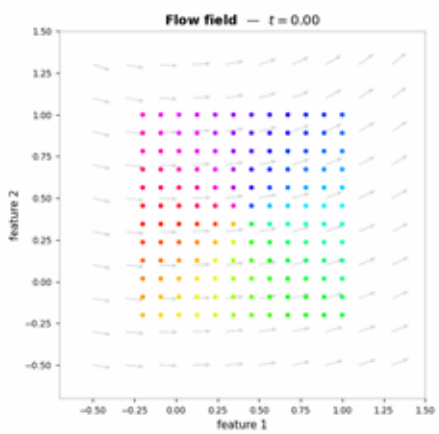
Stochastic Differential Equations

- From ODE to SDE

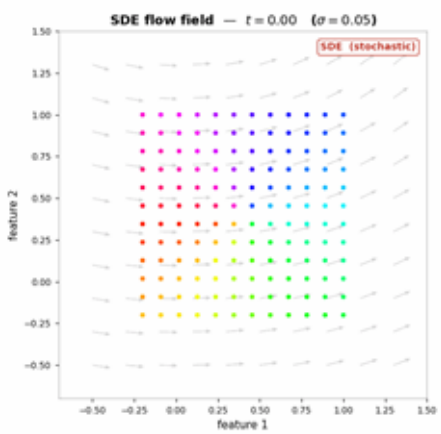
$$\sigma_t \geq 0$$

$$\mathrm{d}X_t = u_t(X_t)\mathrm{d}t + \underline{\sigma_t \mathrm{d}W_t}$$
$$X_0 = x_0$$

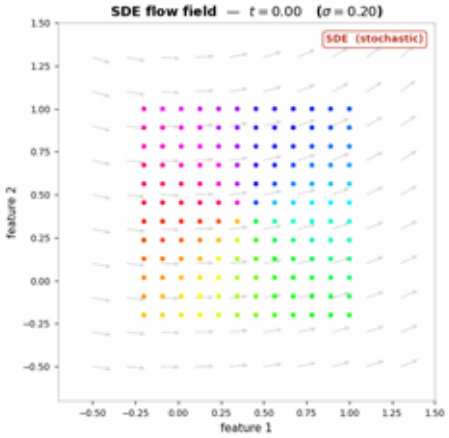
stochastic term by adding Brownian Motion



$\sigma_t=\sigma_0=\sigma$ σ=0 (ode)



σ=0.05



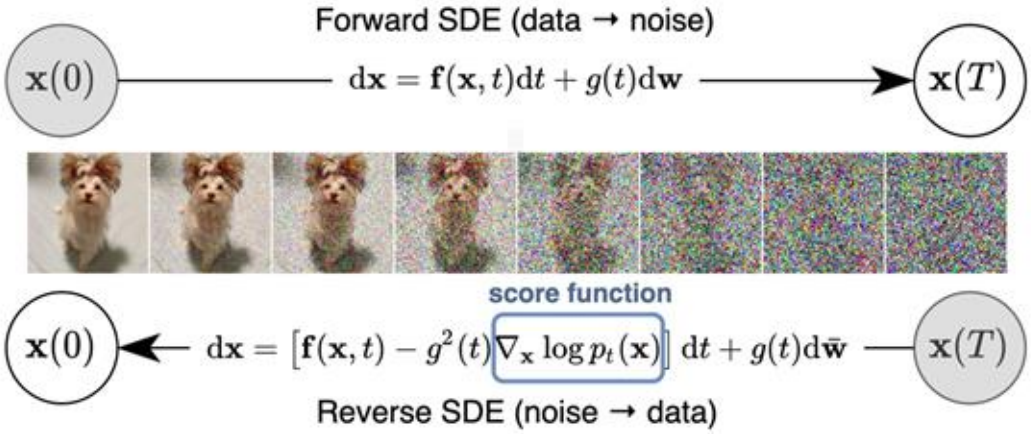
σ=0.20

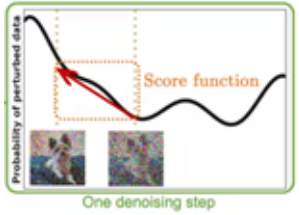
Score-Based Generative Model

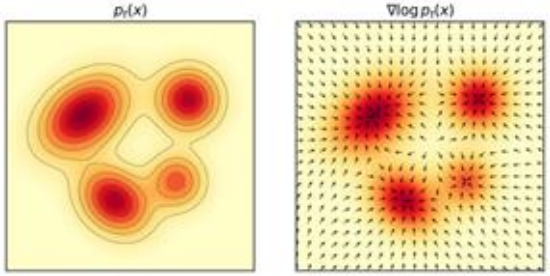
Score-Based Generative Model further extends the discrete process, DDPM, to a **continuous** stochastic process.

A typical stochastic processes are solutions of stochastic differential equations (SDE) including the diffusion process.

Therefore, SDE can be used as a tool to represent diffusion process in a continuous space.







Score-Based Generative Model

Forward Process is represented as SDE

- **Drift coefficient**: Metaphor as mean of forward process
- **Diffusion coefficient**: Metaphor as standard deviation of forward process
- **Derivative of Brownian motion**: Metaphor as a Gaussian noise

$$\mathrm{d}\mathbf{x} = \boxed{\mathbf{f}(\mathbf{x}, t)}\mathrm{d}t + \boxed{g(t)}\boxed{\mathrm{d}\mathbf{w}}$$

Drift coefficient Diffusion coefficient Derivative of Brownian motion

Benefits of formulating through SDE

- Flexibility to design diffusion process through adjusting the drift and diffusion coefficient
- Formulate diffusion process in the continuous space

Score-Based Generative Model

$$\mathrm{d}\mathbf{x} = \mathbf{f}(\mathbf{x}, t)\mathrm{d}t + g(t)\mathrm{d}\mathbf{w}$$

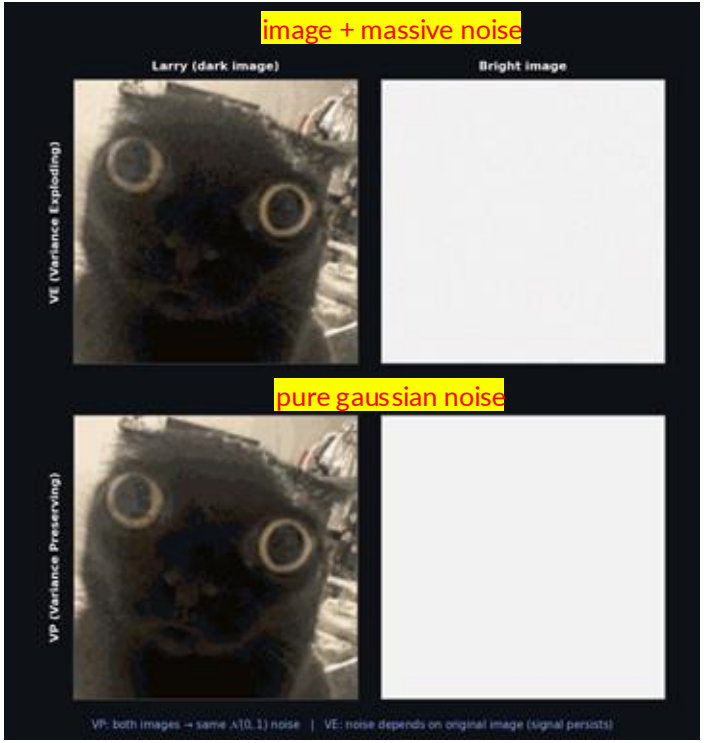
Variance Exploding (VE) and Variance Preserving (VP) SDEs

- **Variance Exploding:** $\mathrm{d}\mathbf{x} = \sqrt{\frac{\mathrm{d}[\sigma^2(t)]}{\mathrm{d}t}}\mathrm{d}\mathbf{w}$

  VE depends on the original data $\mathcal{N}(\mathbf{x}_0, \sigma_{\max}^2 \mathbf{I}_d))$
  Starting the reverse process requires an approximation.

- **Variance Preserving:** $\mathrm{d}\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\mathbf{w}$

  VP converges to pure standard Gaussian noise $\mathcal{N}(0, \mathbf{I}_d)$
  The Advantage: Sampling from pure Gaussian noise to start the reverse process is exact, easy, and completely independent of the original dataset.



Score-Based Generative Model

The connection between DDPM and Score-Based generative model

According to DDPM, forward process: $x_i = \sqrt{1-\beta_i}x_{i-1} + \sqrt{\beta_i}\epsilon_{i-1} \qquad i = 1, \cdots, N$

As N → Infinity, the forward process will converges to the following SDE.

$$\mathrm{d}\mathbf{x} = -\frac{1}{2}\beta(t)\mathbf{x}\mathrm{d}t + \sqrt{\beta(t)}\mathrm{d}\mathbf{w}$$

This proves that DDPM and Score-Based generative model are equivalent.

The difference between these two is a tool to formulate different diffusion processes.

- **DDPM**: Design a diffusion process through ***probabilistic modeling***
- **Score-Based generative model**: Design a diffusion process through ***stochastic differential equation***

Score-Based Generative Model

Any stochastic differential equation has a corresponding reverse SDE:

$$\mathrm{d}\mathbf{x} = [\mathbf{f}(\mathbf{x}, t) - g^2(t)\boxed{\nabla_{\mathbf{x}} \log p_t(\mathbf{x})}]\mathrm{d}t + g(t)\mathrm{d}\mathbf{w}$$

Unknown

In reverse SDE, there is one unknown term, **score function**, and we will train model to predict it. $s_\theta(\mathbf{x}, t) \approx \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$

- An advantage of using a **score function** model is that it lets us disregard the intractable normalization constants.

$$p_\theta(x) = \frac{e^{-f_\theta(x)}}{\boxed{Z_\theta}} \longrightarrow \int p_\theta(x)dx = 1$$

Hard to train

Score-Based Generative Model

Goal: Train a NN to approximate the marginal score: $\underbrace{s_\theta(\mathbf{x}, t)}_{\text{nn output } (\hat{y})} \approx \underbrace{\nabla_{\mathbf{x}} \log p_t(\mathbf{x})}_{\text{label } (y)}$

Unfortunately, The **marginal score** $\nabla_{\mathbf{x}} \log p_t(\mathbf{x})$ is intractable because it requires integrating over the entire data distribution.

The **conditional score** $\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{x}_0 = z)$ assumes a known single clean data point z is perfectly tractable. Through Bayes' Rule, we can derive the following relationship

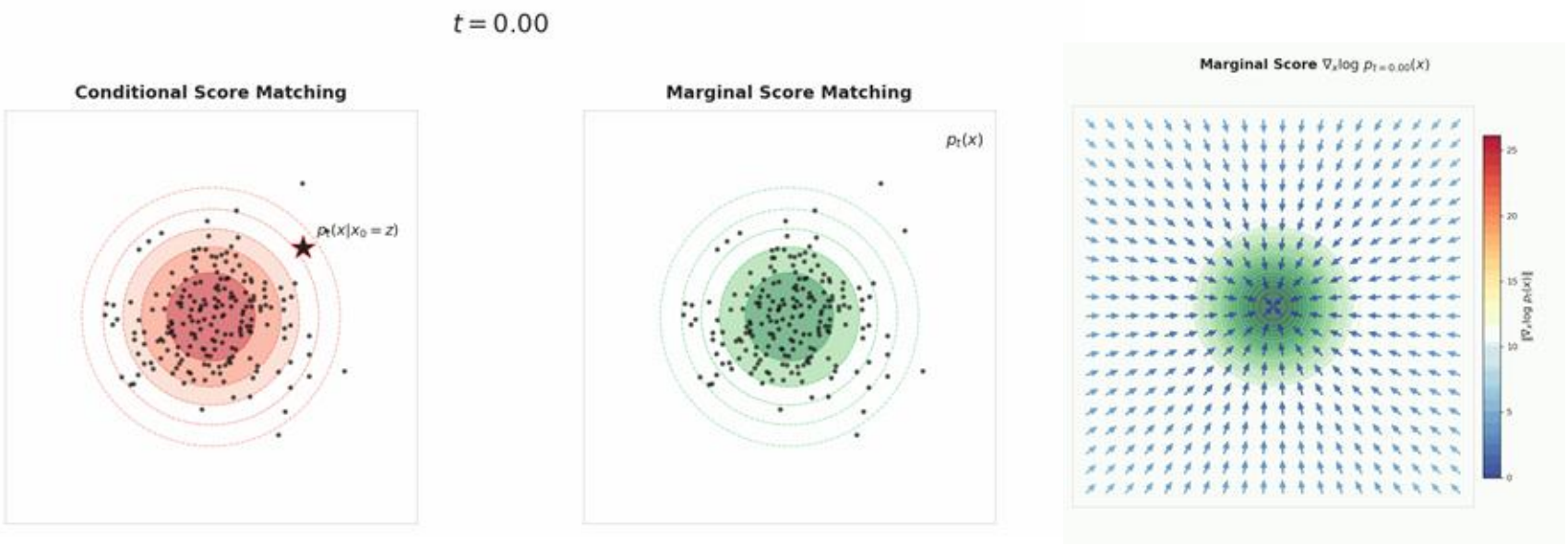
see appendix

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) = \mathbb{E}_{z \sim p_t(\mathbf{x}_0 = z|\mathbf{x})} \left[\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{x}_0 = z) \right]$$

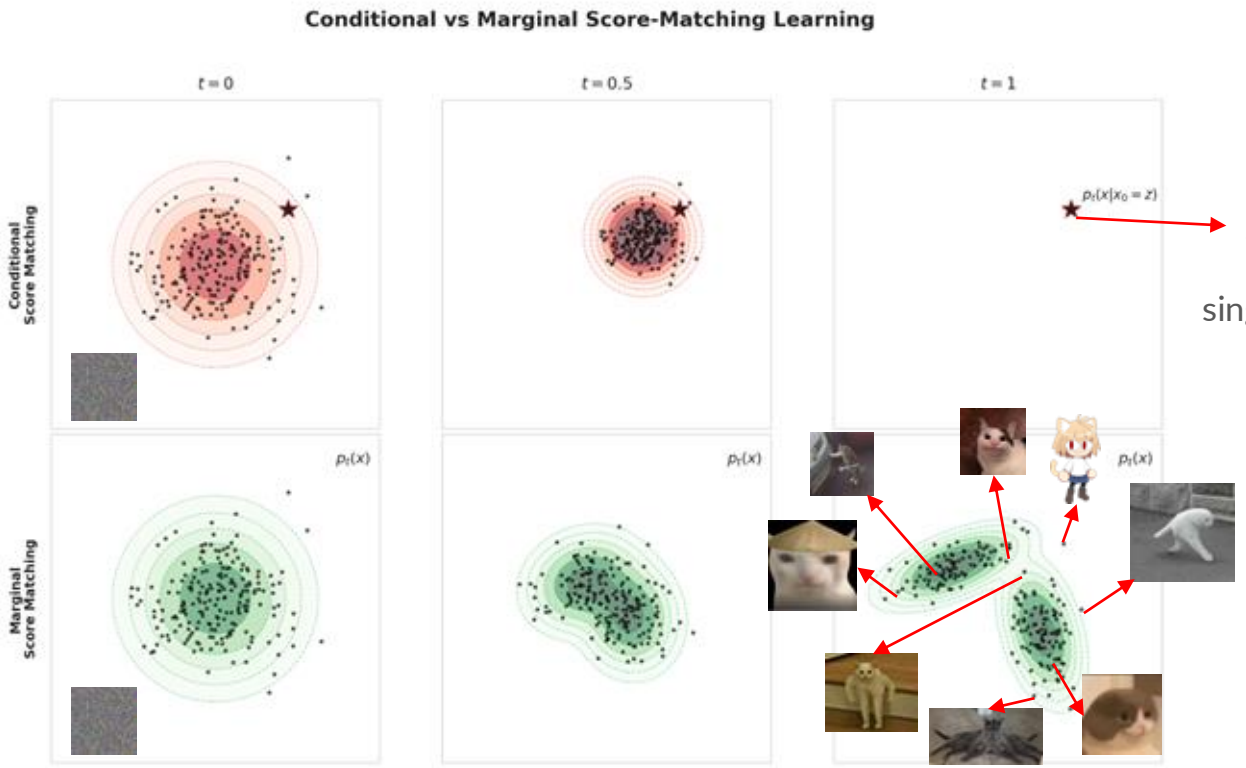
This proves that minimizing the error on the tractable conditional score is mathematically equivalent to learning the intractable marginal score.

“Denoising score matching”

Score-Based Generative Model



Score-Based Generative Model



z=

single data point (conditional)

$p_t(\mathbf{x}|\mathbf{x}_0 = z)$

data distribution (marginal)

$p_t(\mathbf{x})$

Score-Based Generative Model

How to train score function model?

- The objective of *score matching* task is to minimize the following equation.

$$\frac{1}{2}\mathbb{E}_{p_{\text{data}}}[||s_\theta(\mathbf{x}) - \underbrace{\boxed{\nabla_{\mathbf{x}} \log p_{\text{data}}(\mathbf{x})}}_{\text{Unknown}}||_2^2] \Leftrightarrow \mathbb{E}_{p_{\text{data}}(\mathbf{x})}[\text{tr}(\underbrace{\boxed{\nabla_{\mathbf{x}} s_\theta(\mathbf{x})}}_{\text{High dimension}}) + \frac{1}{2}||s_\theta(\mathbf{x})||_2^2]$$

- To circumvent this problem, there are several techniques that can be used
  - **Denoising score matching**: Perturbs the data point **x** with a pre-specified noise

$$\frac{1}{2}\mathbb{E}_{q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})p_{\text{data}}(\mathbf{x})}[||s_\theta(\tilde{\mathbf{x}}) - \nabla_{\tilde{\mathbf{x}}} \log q_\sigma(\tilde{\mathbf{x}}|\mathbf{x})||_2^2]$$

  - **Sliced score matching**: Uses random projections to approximate $\text{tr}(\nabla_{\mathbf{x}} s_\theta(\mathbf{x}))$ in score matching

$$\mathbb{E}_{p_{\mathbf{v}}}\mathbb{E}_{p_{\text{data}}}[\mathbf{v}^{\text{T}}\nabla_{\mathbf{x}} s_\theta(\mathbf{x})\mathbf{v} + \frac{1}{2}||s_\theta(\mathbf{x})||_2^2]$$

Score-Based Generative Model

Using **denoising score matching** technique, the objective is

$$\mathbf{E}_{x_t \sim p_t(x_t)}[||s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)||_2^2]$$

$$\int p_t(\mathbf{x}_t)[||s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)||_2^2]\mathrm{d}\mathbf{x}_t$$

$$\int \int p_t(\mathbf{x}_t|\mathbf{x}_0)p_{\text{data}}(x_0)[||s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)||_2^2]\mathrm{d}\mathbf{x}_0\mathrm{d}\mathbf{x}_t$$

$$\mathbf{E}_{\underline{x_t \sim p_t(x_t|x_0)}, \underline{x_0 \sim p_{\text{data}}(x_0)}}[||s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t)||_2^2]$$

$\boxed{x_t \leftarrow \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon}$ sample from dataset

The probability $\mathbf{x}_t$ given $\mathbf{x}_0$ can be computed by solving the SDE.

Score-Based Generative Model

In practice, we optimize this equation (conditional form):

$$\mathbf{E}_{t\sim\text{Uniform}(0,\ \text{T}),\mathbf{x}_0\sim p_{\text{data}},\mathbf{x}_t\sim p_t(\mathbf{x}_t|\mathbf{x}_0)}[\lambda(t)||s_\theta(\mathbf{x}(t),t)-\nabla_{\mathbf{x}_t}\log p_{0t}(\mathbf{x}(t)|\mathbf{x}(0))||_2^2]$$

where lambda is a weighting function.

- We typically design the weighting function to balance the magnitude of the loss across time.
- When $\lambda(t)=g(t)^2$, the score-based loss function is equal to DDPM loss function.

$$\begin{aligned}
p_{0t}(\mathbf{x}(t)|\mathbf{x}(0)) &= \mathcal{N}(\mu,\sigma^2\mathbf{I}) \\
\nabla_{\mathbf{x}(t)}\log p_{0t}(\mathbf{x}(t)|\mathbf{x}(0)) &= \nabla_{\mathbf{x}(t)}\log\frac{1}{\sqrt{2\pi}\sigma}\exp(-\frac{1}{2}\frac{(x(t)-\mu)^2}{\sigma^2}) \\
&= -\frac{1}{2}\nabla_{\mathbf{x}(t)}\frac{(x(t)-\mu)^2}{\sigma^2} \\
&= -\frac{(x(t)-\mu)}{\sigma^2} \\
&= -\frac{\epsilon}{\sigma}, \qquad \mathbf{x}(t)=\mu+\sigma\epsilon
\end{aligned}$$

Score-Based Generative Model

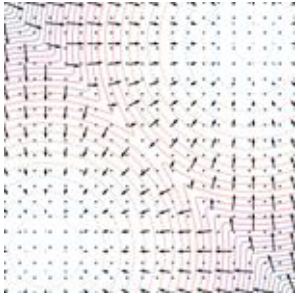
Inference consists of two steps:

1. **Predictor Step**
   a. Sampling $\mathbf{x}(t+\Delta t) \sim p_{t+\Delta t}(\mathbf{x})$ from $\mathbf{x}(t) \sim p_t(\mathbf{x})$ using any numerical SDE solver such as Euler-Maruyama.

$$\begin{aligned} \Delta\mathbf{x} &\leftarrow [\mathbf{f}(\mathbf{x},t) - g^2(t)s_\theta(\mathbf{x},t)]\Delta t + g(t)\sqrt{|\Delta t|}\epsilon_t \\ \mathbf{x} &\leftarrow \mathbf{x} + \Delta\mathbf{x} \\ t &\leftarrow t + \Delta t \end{aligned}$$

1. **Corrector Step**
   a. Refine $\mathbf{x}(t+\Delta t) \sim p_{t+\Delta t}(\mathbf{x})$ from predictor step using MCMC procedure such as Langevin dynamics

$$\mathbf{x}_{t+1} \leftarrow \mathbf{x}_t + s_\theta(\mathbf{x},t)\Delta t + \sqrt{2\Delta t}\epsilon$$



Score-Based Generative Model

Algorithm 3 Training

repeat
  $x_0 \sim q(x_0)$
  $t \sim \text{Uniform}(\{1, \ldots, T\})$
  $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
  $x_t \leftarrow \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon$
  Take gradient descent step on $\nabla_\theta \lambda(t) || s_\theta(\mathbf{x}_t, t) - \nabla_{\mathbf{x}} \log p_t(\mathbf{x}_t) ||_2^2$
until converged

Algorithm 4 Sampling

$x_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$
for $t = T, \ldots, 1$ **do**
  Predictor Step (x_t, t)
  Corrector Step (x_t, t)
end for
return x_0

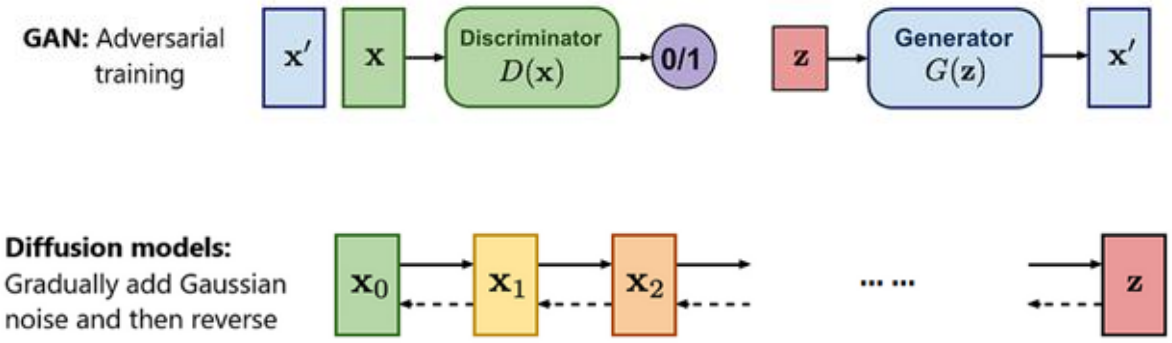
Further research on diffusion

Speed
Architecture

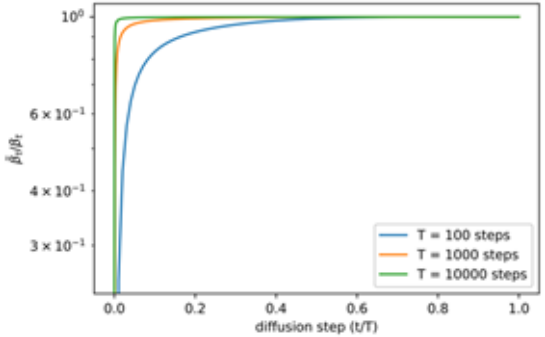
Open Research Question in Diffusion

The main problem with diffusion model is slow Inference time.

- It requires several diffusion step to obtain high-quality sample. Conversely other approach such as GAN only take one inference step.
- The diffusion model cannot be parallelized due to its Markovian nature. (Cannot utilize GPU effectively)



How can we shorten the diffusion steps?



During training process, we train diffusion model to perform N diffusion step.

Is it possible to perform only S step (S < N) when inference? (with a fixed variance)

- No, it is inappropriate to do so. As mentioned earlier, the gap between the upperbound and lowerbound becomes larger and larger when the diffusion step is small.

Therefore, in the Improved Denoising Diffusion Probabilistic Model, they proposed **learning the variance of the reverse process**. Moreover, the variance was parameterized as an interpolation between the upperbound and lowerbound in the log domain, due to the infinitesimal range of reasonable values.

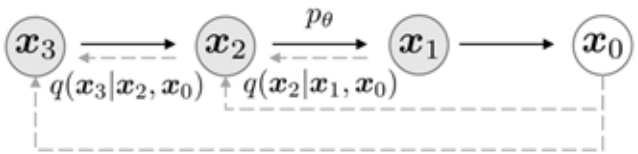
$$\Sigma_\theta(x_t, t) = \exp(v_\theta \log \beta_t + (1 - v_\theta) \log \tilde{\beta}_t)$$

Unweighted loss (vanilla version) is still a main source for mean and the weighted version is used to train variance specifically.

$$L_{\text{hybrid}} = \underbrace{L_{\text{simple}}}_{\text{unweighted loss}} + \lambda \underbrace{L_{\text{vlb}}}_{\text{weighted loss}}$$

What about the sampling method? Can we change?

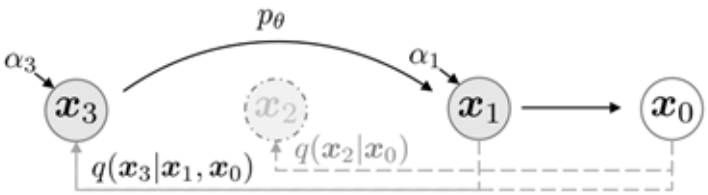
In Denoising Diffusion Implicit Model, the forward process is modified to be non-Markovian.



As a result, DDIM introduces a **new sampling method** while retaining the same loss as DDPM (use as pre-trained weights).

This sampling method enables the selection of only a subset of the sequence to expedite the inference process.

$$q_\sigma(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \sqrt{\alpha_{t-1}}x_0 + \sqrt{1-\alpha_{t-1}-\sigma_t^2} \cdot \frac{x_t - \sqrt{\alpha_t}x_0}{\sqrt{1-\alpha_t}}, \sigma_t^2\mathbf{I})$$



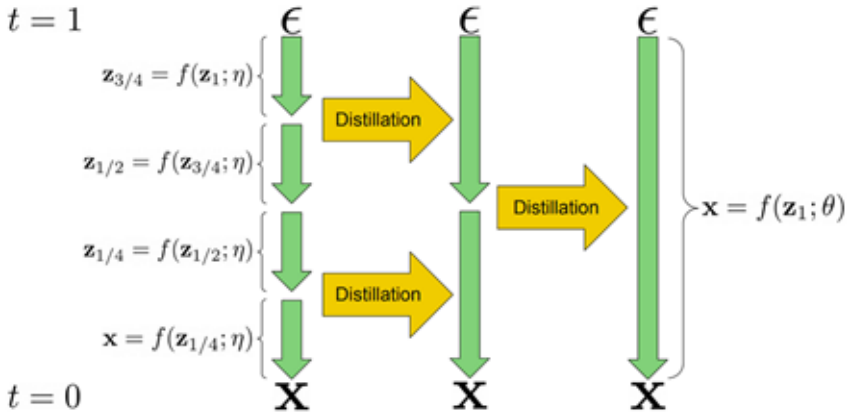
Others method like this: DPM solver

https://arxiv.org/abs/2010.02502 https://arxiv.org/abs/2206.00927

Knowledge Distillation

The paper propose a method to transfer knowledge from teacher model to student model so that student require fewer diffusion step than teacher model.

1. Training a teacher model similar to vanilla diffusion model
2. Transferring knowledge from teacher model to student model
    a. One step in student model equivalent to two steps in teacher model.
3. Set student model as teacher model and go back to step 2.



https://arxiv.org/abs/2202.00512 **Progressive Distillation for Fast Sampling of Diffusion Models** 2022

On Distillation of Guided Diffusion Models

Extend the idea of knowledge distillation from the previous slide on the guided diffusion models

1. Stage-one distillation
    a. Transfer the knowledge from the classifier-free diffusion model to a student model, enabling the student model to predict interpolated noise in a single pass.

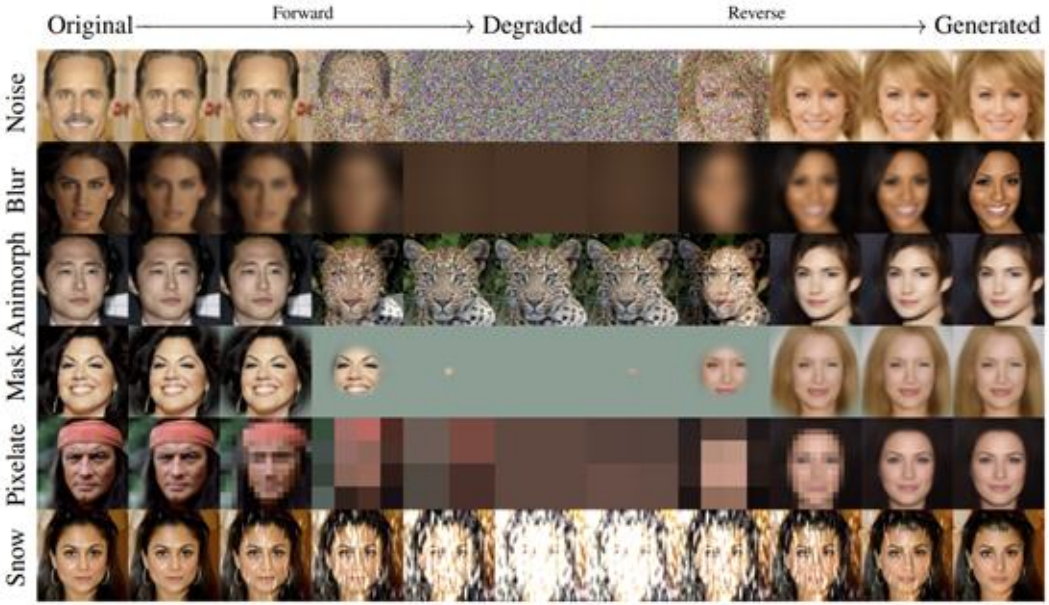
$$\hat{\mathbf{x}}^{w}_{c,\theta}(\mathbf{z}_t) = (1+w)\hat{\mathbf{x}}_{c,\theta}(\mathbf{z}_t) - w\hat{\mathbf{x}}_{\theta}(\mathbf{z}_t)$$

$$\mathbb{E}_{w\sim U[w_{\min},w_{\max}],t\sim U[0,1],\mathbf{x}\sim p_{\text{data}}(\mathbf{x})}\left[w(\lambda_t)||\hat{\mathbf{x}}_{\eta_1}(\mathbf{z}_t,w) - \hat{\mathbf{x}}^{w}_{c,\theta}(\mathbf{z}_t)||_2^2\right]$$

1. Stage-two distillation
    a. Use the student model from step 1 as a teacher model and follow the same procedure as in the previous slide.

Does it need to be Gaussian Noise?

In the cold diffusion paper, they showed that it is not necessary to use Gaussian noise but any image transformation. This raises questions about the limits of our theoretical understanding of diffusion models.



https://arxiv.org/abs/2208.09392 2022 Cold Diffusion: Inverting Arbitrary Image Transforms Without Noise

DiT (e.g. Attention is all you need ver. Diffusion)

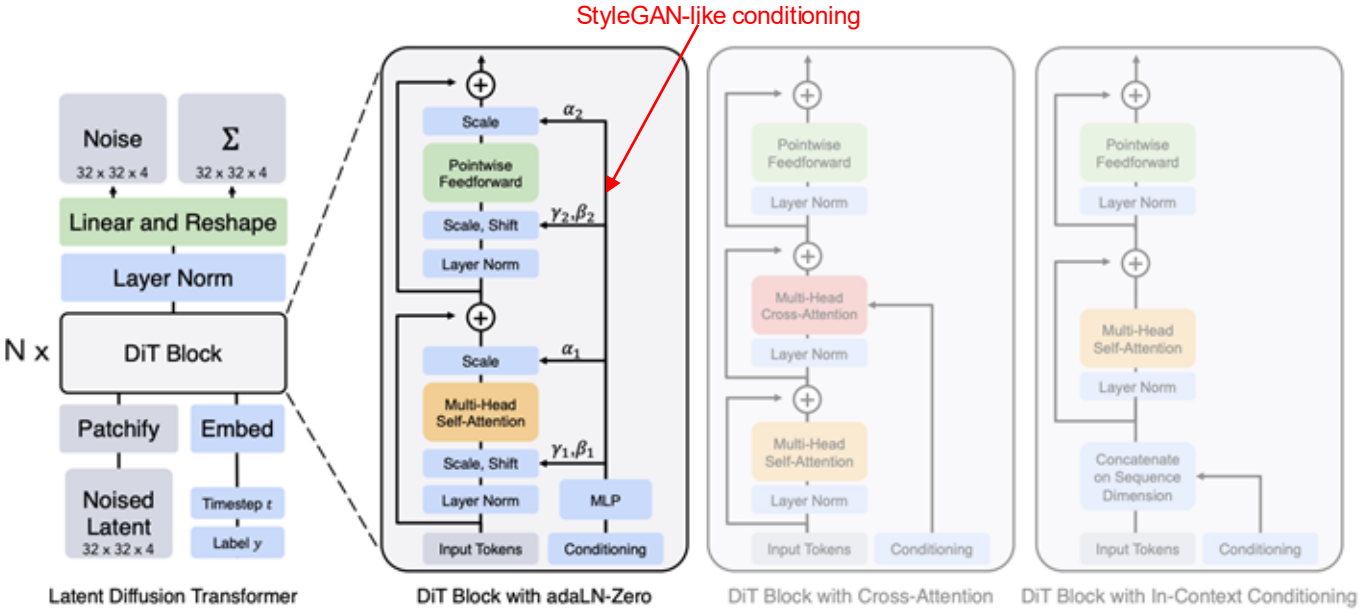


Figure 3. **The Diffusion Transformer (DiT) architecture.** *Left:* We train conditional latent DiT models. The input latent is decomposed into patches and processed by several DiT blocks. *Right:* Details of our DiT blocks. We experiment with variants of standard transformer blocks that incorporate conditioning via adaptive layer norm, cross-attention and extra input tokens. Adaptive layer norm works best.

https://arxiv.org/abs/2212.09748 Scalable Diffusion Models with Transformers (DiT) 2023

Additional Interesting Paper

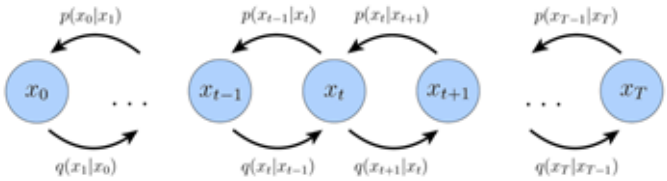
1. Variational Diffusion Models https://arxiv.org/abs/2107.00630 (How to learn noise scheduler?)
2. Lookahead Diffusion Probabilistic Models for Refining Mean Estimation (Refining the predicted mean)
3. Learning Fast Samplers for Diffusion Models by Differentiating Through Sample Quality (Finding General Family of DDIM)
4. Speech Generation and Speech Enhancement:
    a. WaveGrad2 Iterative Refinement for Text-to-Speech Synthesis
    b. WaveFit An Iterative and Non-autoregressive Neural vocoder Based on Fix-Point Iteration
    c. DiffWave A Versatile Diffusion Model for Audio Synthesis
    d. Speech Enhancement and Dereverberation with Diffusion-based Generative Models

Homework

Tutorial
Homework

Summary

- Diffusion vs Flow
  - They are quite similar (and can be shown to be the same thing under certain settings) except that flow requires invertible layers
    - Flow -> exact max likelihood
    - Diffusion -> solve via ELBO
  - Recent view of flow models
    - Flow matching https://arxiv.org/abs/2210.02747
      - View diffusion as a special case of flow
- Serveral formulation of diffusions
  - DDPM vs Score-based



Appendix
Splitting diffusion loss into 3 terms

1.2.1 Loss Function

As a result, $\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_{1:T}|x_0) - \log p(x_{0:T})]$ is optimized instead of $\mathbf{E}_{x_0\sim q(x_0)}[-\log p(x_0)]$ which is tractable to compute. We further examine this equation and split it into 3 terms.
Consider $\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_{1:T}|x_0) - \log p(x_{0:T})]$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_{1:T}|x_0) - \log p(x_{0:T})]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \prod_{t=1}^{T} q(x_t|x_{t-1}) - \log p(x_T) \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_1|x_0) + \log \prod_{t=2}^{T} q(x_t|x_{t-1}) - \log p(x_T) \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

Forward process is Markov process, then $q(x_t|x_{t-1}) = q(x_t|x_{t-1}, x_0)$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_1|x_0) + \log \prod_{t=2}^{T} q(x_t|x_{t-1}, x_0) - \log p(x_T) \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log q(x_1|x_0) - \log p(x_T) + \log \prod_{t=2}^{T} q(x_t|x_{t-1}, x_0) - \log \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} q(x_t|x_{t-1}, x_0) - \log \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} \frac{q(x_t, x_{t-1}, x_0)}{q(x_{t-1}, x_0)} - \log \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)q(x_t, x_0)}{q(x_{t-1}, x_0)} - \log \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)q(x_t, x_0)}{q(x_{t-1}, x_0)} - \log \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)q(x_t, x_0)}{q(x_{t-1}, x_0)} - \log \prod_{t=1}^{T} p(x_{t-1}|x_t)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} q(x_{t-1}|x_t, x_0) - \log \prod_{t=1}^{T} p(x_{t-1}|x_t) + \log \prod_{t=2}^{T} \frac{\cancelto{q(x_T,x_0)}{q(x_t, x_0)}}{\cancelto{q(x_1,x_0)}{q(x_{t-1}, x_0)}}]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)}{p(x_{t-1}|x_t)} - \log p(x_0|x_1) + \log \frac{q(x_T|x_0)}{q(x_1|x_0)}]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_1|x_0)q(x_T|x_0)}{p(x_T)q(x_1|x_0)} + \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)}{p(x_{t-1}|x_t)} - \log p(x_0|x_1)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\log \frac{q(x_T|x_0)}{p(x_T)} + \log \prod_{t=2}^{T} \frac{q(x_{t-1}|x_t, x_0)}{p(x_{t-1}|x_t)} - \log p(x_0|x_1)]$$

$$\mathbf{E}_{x_{0:T}\sim q(x_{0:T})}[\underbrace{D_{\mathrm{KL}}(q(x_T|x_0)||p(x_T))}_{L_T} + \sum_{t=2}^{T} \underbrace{D_{\mathrm{KL}}(q(x_{t-1}|x_t, x_0)||p(x_{t-1}|x_t))}_{L_{t-1}} - \underbrace{\log p(x_0|x_1)}_{L_0}]$$

Appendix
Getting $q(x_{t-1}|x_t, x_0)$

From $q(x_t|x_{t-1}) = \mathcal{N}(\sqrt{\alpha_t}x_{t-1}, (1-\alpha_t)\mathbf{I})$ and $q(x_t|x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t}x_0, (1-\bar{\alpha}_t)\mathbf{I})$, and let $\beta_t = 1 - \alpha_t$

$$q(x_{t-1}|x_t, x_0) = \frac{1}{\sqrt{\beta_t}\sqrt{2\pi}} \frac{\sqrt{1-\bar{\alpha}_t}}{\sqrt{1-\bar{\alpha}_{t-1}}} \exp(-\frac{1}{2}(\underbrace{\frac{(x_t - \sqrt{\alpha_t}x_{t-1})^2}{\beta_t} \cdots}_{\text{Consider this term}})$$

Consider the term inside the exponential function.

$$\frac{(x_t - \sqrt{\alpha_t}x_{t-1})^2}{\beta_t} + \frac{(x_{t-1} - \sqrt{\bar{\alpha}}_{t-1}x_0)^2}{1-\bar{\alpha}_{t-1}} - \frac{(x_t - \sqrt{\bar{\alpha}}_t x_0)^2}{1-\bar{\alpha}_t}$$

$$\underbrace{(\frac{\alpha_t}{\beta_t} + \frac{1}{1-\bar{\alpha}_{t-1}})}_{a} x_{t-1}^2 - \underbrace{(\frac{2\sqrt{\alpha_t}}{\beta_t}x_t + \frac{2\sqrt{\bar{\alpha}}_{t-1}}{1-\bar{\alpha}_{t-1}})}_{b} x_{t-1} + \underbrace{\frac{x_t^2}{\beta_t} + \frac{\bar{\alpha}_{t-1}x_0^2}{1-\bar{\alpha}_{t-1}} - \frac{(x_t - \sqrt{\bar{\alpha}}_t x_0)^2}{1-\bar{\alpha}_t}}_{c}$$

From $ax^2 + bx + c = a(x + \frac{b}{2a})^2 + \frac{4ac-b^2}{4a}$

$$\frac{(x_{t-1} - (\frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}}_{t-1}\beta_t}{1-\bar{\alpha}_t}x_0))^2}{\frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}}$$

Therefore $q(x_{t-1}|x_t, x_0) = \mathcal{N}(x_{t-1}; \tilde{\mu}_t, \tilde{\beta}_t\mathbf{I})$

$$\tilde{\mu}_t = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}x_t + \frac{\sqrt{\bar{\alpha}}_{t-1}\beta_t}{1-\bar{\alpha}_t}x_0$$

$$\tilde{\beta}_t = \frac{\beta_t(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}$$

Appendix
Marginal score function

$$
\begin{aligned}
\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) &= \frac{\nabla_{\mathbf{x}} p_t(\mathbf{x})}{p_t(\mathbf{x})} \\
&= \frac{1}{p_t(\mathbf{x})} \nabla_{\mathbf{x}} \int p_t(\mathbf{x}|\mathbf{x}_0 = z) p_{\text{data}}(z) \mathrm{d}z \\
&= \frac{1}{p_t(\mathbf{x})} \int \nabla_{\mathbf{x}} p_t(\mathbf{x}|\mathbf{x}_0 = z) p_{\text{data}}(z) \mathrm{d}z \\
&= \int (\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{x}_0 = z)) \frac{p_t(\mathbf{x}|\mathbf{x}_0 = z) p_{\text{data}}(z)}{p_t(\mathbf{x})} \mathrm{d}z \\
&= \int (\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{x}_0 = z)) p_t(\mathbf{x}_0 = z|\mathbf{x}) \mathrm{d}z \\
&= \mathbb{E}_{z \sim p_t(\mathbf{x}_0 = z|\mathbf{x})} \left[\nabla_{\mathbf{x}} \log p_t(\mathbf{x}|\mathbf{x}_0 = z)) \right]
\end{aligned}
$$